

Stage 1 Tutorial Activity

Why Software Engineering Still Matters | Week 4 | Part A (Tutorial, 1.5 marks)

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Tutorial group	
Submission	Week 4 - Stage 1

Activity 1 - Think-Pair-Share

Question: If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Real **problem and stakeholders understanding**. AI responds to the prompt it is given, not the need behind it. Requirements are elicited and validated by an engineer (for SmartCare: who books, who cancels, what 'duplicate' means) to ensure that the right thing is built.
2. Design judgment and compromises. Code that runs is not necessarily maintainable, correct for edge cases or appropriately scoped. Engineers decide structure, responsibilities, data integrity rules and what not to build (e.g. no hospital system).
3. Verification, accountability and ethics. Someone has to test, debug and review generated code, safeguard patient privacy, and be responsible for it when it fails. Generated code may appear confident, but be wrong, insecure or out of scope, and must be trusted with evidence.

Activity 2 - Is This Software Engineering?

Scenario	Programming?	Software engineering?	Why?
A. Student writes a 50-line Python calculator	Yes	Minimal / not really	It's programming: one author, one user, a small known problem. There are no stakeholders, requirements negotiation, long-term maintenance or team coordination, so engineering practices add little.
B. Team develops a payroll system for 5,000 employees	Yes	Yes - clearly	Many stakeholders (staff, HR, finance, auditors), legal rules, high cost of error, teamwork, requirements, design, testing, deployment and years of maintenance. Programming is just one of the activities.
C. AI assistant generates a simple appointment app from one prompt	Yes - the AI did the programming	Not by itself; becomes SE only when humans add it	A prompt is not a specification. If there is no engineering process, then there is no scope, no testing, no reviewing, and no ownership. If there is no scope, there is no testing, there is no reviewing, and there is no ownership, then there is no engineering process. When validated against the client's needs and verified, the same output is called software engineering.

Activity 3 - SmartCare Problem Analysis

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Receptionist / administrative staff	Quick patient search; book and cancel appointments without duplicates; view appointments status clearly and consistently.
Practitioners (GPs)	Check their own schedule and availability; assume the appointments listed are accurate and up-to-date.
Patients	No double booking or lost appointments, reliable bookings, records are found quickly and kept private.
Clinic manager / owner	Good appointment history and simple operational reports; a small clinic system that is easy to manage and affordable.
IT support / maintainer (junior engineer)	Clear, easy to understand, testable code that can be modified safely as the clinic's needs become more clear.
Privacy regulator (external)	Personal information used for health purposes in accordance with Australian privacy legislation (e.g. Privacy Act 1988).

Task 2 - Identify current problems

1. Duplicate appointment bookings due to no cross-referencing between spreadsheets and paper records
2. Difficulty locating patient records spread over spreadsheets and paper files.
3. **Appointment status is inconsistent (e.g. an appointment is cancelled but still shows as booked), staff can't rely on the schedule.**
4. **Practitioner availability is limited and there is no reliable history of practitioner appointments, and manual cancellation and inability to produce basic reports.**

Task 3 - Ask client questions

1. What is meant by 'duplicate booking' - is it the same practitioner at the same time, the same patient twice, or both?
2. Who will be using the system (receptionists, GPs, manager) and do different people require different permissions?
3. What patient information should be kept and what privacy/security requirements should be adhered to?
4. What is the number of patients, practitioners and appointments per day in the clinic, and what is the duration of a typical appointment?
5. Who can cancel an appointment, how late, does a reason need to be recorded, is the history kept?

Activity 4 - Critique an AI Response

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes - duplicate bookings and 'manual cancellation; manage patients, practitioners and appointments	Yes	Accept
Facial recognition login	None - No login or biometric requirement mentioned	No	Reject (privacy risk, cost, no evidence)
AI diagnosis recommendations	None - clinic requested administration, not clinical advice	No	Reject (clinical safety/regulatory risk)
Patient search	Yes - difficulty finding patient records	Yes	Accept
Online payment	None - no mention of billing	No	Reject for v1; ask the client if needed later
Practitioner schedule view	Yes - limited visibility of practitioner availability	Yes	Accept (detail provisional)

Suggestion	Client evidence?	In scope?	Decision
Insurance processing	None	No	Reject (complex, outside 'not a hospital system')
Treatment-plan generation	None - clinical, not administrative	No	Reject (clinical risk, out of scope)

Pattern: the three accepted features trace directly to the client's listed problems; the five rejected ones are plausible-sounding AI additions with no client evidence.

Exit question

There is one activity that a software engineer has to do that can't be fully automated by AI: validate requirements with the real stakeholders (and then validate that the software built meets those requirements). AI does not have access to the actual rules, risks or priorities of SmartCare, and can only generate plausible text; a human must verify what the clinic actually requires and be responsible for the outcome.